

Complements of Operational Research

4.3 Replacement theory: sudden failures and group replacement

José Firmino Aguilar Madeira

Equipment that fails suddenly

Lamps, valves, pumps, sensors, small motors: they work until they fail **for good**, with no warning. When there are many identical units, there are two policies:

- **Individual replacement:** each unit is replaced when it fails.
- **Group replacement:** **everything** is replaced every t periods (even what still works); between group replacements, whatever fails is replaced individually.

Why group replacement may pay off

Replacing in bulk, in a planned way, costs much less per unit: one trip, one aerial work platform, outside working hours, no unscheduled stoppages.

Question: does it pay to replace everything?
How often?

The example: 800 light fittings in a warehouse

A logistics warehouse has $n = 800$ LED light fittings at a height of 12 m.

- individual replacement: $c_i = 25$ € (fitting, platform, technician);
- group replacement: $c_g = 6$ € per fitting.

Mean life

$$\begin{aligned}\bar{t} &= \sum_t t p_t = \\ &= 1(0.03) + 2(0.05) + \cdots + 6(0.25) \\ &= 4.49 \text{ semesters (about 2 years and 3 months).}\end{aligned}$$

Cumulative probability $P(t)$ that a new fitting has failed by the end of each semester, and probability p_t that it fails **in** semester t :

Semester t	1	2	3	4	5	6
$P(t)$	0.03	0.08	0.20	0.45	0.75	1
$p_t = P(t) - P(t-1)$	0.03	0.05	0.12	0.25	0.30	0.25

In the long run, each position gets a new fitting, on average, every $\bar{t}$ semesters. Hence, on average,

$$\frac{n}{\bar{t}} = \frac{800}{4.49} = 178.2 \text{ fittings fail per semester.}$$

Cost of the individual policy

$$K_{\text{ind}} = \frac{n}{\bar{t}} c_i = 178.2 \times 25 = \mathbf{4454.34} \text{ € per semester}$$

But at the start the failures are not 178 per semester: the fittings are all new and rarely fail; then many fail at the same time; the replacements get out of step and the number settles down.

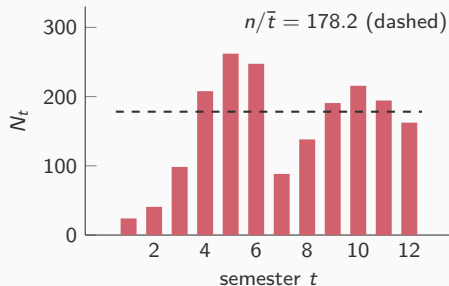
To assess the group policy we need the expected number of failures **in each** semester, N_t .

Expected failures in each semester

The failures in semester t come from the original fittings and from the replacements installed in earlier semesters:

$$N_t = n p_t + N_1 p_{t-1} + \cdots + N_{t-1} p_1.$$

- $N_1 = 800(0.03) = 24$
- $N_2 = 800(0.05) + 24(0.03) = 40.72$
- $N_3 = 800(0.12) + 24(0.05) + 40.72(0.03) = 98.42$
- $N_4 = 207.87, N_5 = 262.04, N_6 = 247.45, \dots$



The failures oscillate and approach $n/\bar{t}$.

Group policy every t semesters

All are replaced at the end of semester t (cost $n c_g = 4800$); those that fail before then are replaced individually.

$$K(t) = \frac{c_i \sum_{k=1}^t N_k + n c_g}{t}$$

t	N_t	$\sum N_k$	$25 \sum N_k$	total	$K(t)$
1	24.00	24.00	600.0	5400.0	5400.0
2	40.72	64.72	1618.0	6418.0	3209.0
3	98.42	163.14	4078.5	8878.5	2959.5
4	207.87	371.01	9275.3	14075.3	3518.8
5	262.04	633.05	15826.3	20626.3	4125.3
6	247.45	880.50	22012.5	26812.5	4468.8

Decision

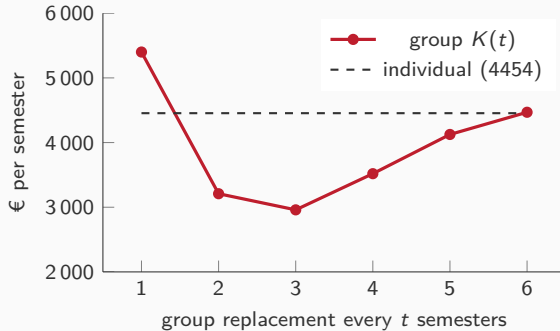
Group replacement every 3 semesters: **2959.5 €/semester**, against 4454.3 € for the individual policy.

Saving: 1494.8 € per semester (−34%), about 3000 € per year.

Marginal reading (in this model): group replacement every 3 semesters is optimal because the individual cost of the next semester, $c_i N_{t+1}$, already exceeds $K(t)$: $25 \times 207.87 = 5197 > 2959.5$, whereas $25 \times 98.42 = 2461 < K(2) = 3209$.

This is a check, not a general block-replacement rule: the decision rests on the table of $K(t)$.

Individual versus group



- Replacing everything every semester is expensive: many good fittings are thrown away.
- Waiting too long is too: individual failures (at 25 €) pile up.
- Between 2 and 3 semesters the group policy is clearly better.

Sensitivity: with $c_g = 10$ € group replacement still wins ($K(3) = 4026$); with $c_g = 12$ € it no longer does ($K(3) = 4560 > 4454$). The result depends heavily on the difference between c_i and c_g .

Confirming by simulation

```
import numpy as np
P = [0.03, 0.08, 0.20, 0.45, 0.75, 1.0]
p = np.diff([0] + P); vida = np.arange(1, 7)
rng = np.random.default_rng(1)
n, T, corridas = 800, 8, 4000
F = np.zeros(T)
for _ in range(corridas):
    falha = rng.choice(vida, size=n, p=p) # semester of 1st failure
    for t in range(1, T + 1):
        f = falha == t
        F[t-1] += f.sum()
    falha[f] = t + rng.choice(vida, size=f.sum(), p=p)
print(np.round(F / corridas, 1))
```

t	N_t (formula)	simulation
1	24.0	24.0
2	40.7	40.7
3	98.4	98.5
4	207.9	207.7
5	262.0	262.2
6	247.4	247.3
7	88.3	88.4
8	138.1	138.3

4000 simulations of 800 fittings. Simulation does without the formula and allows more realistic rules (for example, also replacing those more than 2 years old).

Example 5 — light bulbs

An installation has 1000 light bulbs. The individual replacement cost is 3 € per bulb and the group replacement cost is 1 € per bulb. The cumulative failure probability by the end of each week is:

Week	1	2	3	4	5
$P(t)$	0.10	0.25	0.50	0.70	1.00

- a) What is the mean life of a bulb? b) What is the weekly cost of individual replacement?
c) What is the best group policy? Does it pay off compared with individual replacement?

Exercise sheet 4 — Exercise 4 (homework)

Optimising maintenance strategies in a chemical plant (homework)

A chemical plant has 1000 centrifugal pumps, each essential for continuous operation. **Preventive maintenance** can be done at any time, at 2.5 m.u. per pump, and restores the pump to its initial condition. If a pump breaks down, **corrective maintenance** costs 2.5 m.u. per pump plus 7.5 m.u. for the unscheduled stoppage. The failure probability as a function of the time since the last maintenance is:

Months since the last maintenance	1	2	3	4	≥ 5
(Cumulative) failure probability	0.10	0.25	0.50	0.80	1

1. What is the optimal frequency of preventive maintenance?
2. What is the expected total annual cost of the optimal strategy?

Submission: by email to the lecturer, by the deadline given in class. All exercises: **Exercise sheet 4**.

KEY TAKEAWAYS — 4.3

1. $p_t = P(t) - P(t - 1)$; mean life $\bar{t} = \sum t p_t$; individual: $K_{\text{ind}} = (n/\bar{t}) c_i$ per period.
2. Expected failures: $N_t = n p_t + \sum_{k=1}^{t-1} N_k p_{t-k}$.
3. Group every t : $K(t) = (c_i \sum_{k \leq t} N_k + n c_g)/t$; choose the t with the lowest K and compare with K_{ind} .
4. For annual costs, multiply by the number of periods per year.
5. Simulation confirms the calculations and allows more realistic policies.

End of chapter 4. Next: 5 Dynamic programming.

References Full list: *Course bibliography*.

- Kasana, H. S., Kumar, K. D., *Introductory Operations Research*, Springer, 2004 — ch. 11 (*group replacement*).
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017.
- Jardine, A. K. S., Tsang, A. H. C., *Maintenance, Replacement, and Reliability*, 2nd ed., CRC Press, 2013 — block replacement policies.
- Barlow, R. E., Proschan, F., *Mathematical Theory of Reliability*, SIAM, 1996 — renewal theory.

Resources

- Exercise sheet 4. `code/python/04_replacement/04_3_group_replacement/grupo.py` (includes the simulation).